lab 6:
User defined function

Objective
To be familiar with user defined function and recursive function in C.

program#1
PROBLEM ANALYSIS
To develop a c program to find sum of two numbers using user defined function, with arguments and return type, we take two numbers as input from user and use in datatype.

ALGORITHM
1) Start
2) Declare variable
3) Enter variable in main ()
4) Read a and b a call sum (a,b)
5) Display sum and End main ()
6) Define function sum ()
7) calculate sum.
8) Return s and End sum ()
9) Stop

C-program
```
#include <stdio.h>
int sum (int,int);
int main ()
{
int a,b,s;
printf ("Enter two numbers:\n");
scanf ("%d %d", &a,&b);
s=sum (a,b);
printf("sum of %d & %d : %d", a, b,s);
return 0;
}
int sum (int x, int y)
{
```

```
int s;
s=x+y;
return s;
}
```

Output

Enter two numbers 2 5

The sum of 2 & 5 = 7

RESULT

The sum of two no. were displayed successfully.

* With arguments and no-returns type

```
#include <stdio.h>
void sum(int, int);
int main ()
{
int a,b;
printf ("Enter two numbers: \n");
scanf ("%d %d",& a & b);
sum (a,b);
return 0;
}
void sum (int c, int d)
{
int s;
s= c+d;
printf ("The sum is %d", s);
}
```

RESULT

Two no. were added successfully.

* Without arguments and return type

```
#include <stdio.h>
int sum ()
int main ()
{
int s;
```

```
s= sum ();
printf ("The sum is %d ", s);
return 0;
}
int sum()
{
int a, b, s;
printf ("Enter two no: s \n");
scanf ("%d %d", &a, &b);
s= a+b;
return s;
}
```

Output

Enter two no: 2 7

The sum is : 9

RESULT

Two no. were added successfully

* Without arguments and no-return type:

```
# include <stdio.h>
void sum ();
int main ()
{
int s;
sum ();
return 0;
}
void sum()
{
int a, b, s;
printf ("Enter two no: \n");
scanf ("%d %d", &a, &b);
s= a+b;
printf ("Sum of %d and %d is %d", a, b, s);
}
```

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Output

Enter two no: 20 80

The sum of 20 and 80 is 100.

RESUIT

Two no. were added successfully.

PROGRAM #3

PROBIEM ANAIYSIS

To write a program in C to check whether the no. given is prime or not using user-defined function we can create a function prime () or else and through main check.

AIGORITHM

1. Start
2) CREATE Function PRIME ()
3) Initialize i:=2 to i<n/2 and check condition for prime
4) If (n%i==0), then its prime or not.
5) CREATE main ()
6) Read no. and call the prime ()
7) Stop

C-program

```
#include <stdio.h>
int prime (int);
int main ()
{
printf ("Enter n: \n");
scanf ("%d", &n);
if (isprime (n)==1);
{
printf ("%d is prime", n);
}
else
{
printf ("%d is not prime", n);
}
return 0;
}
```

```
int isprime (int n)
{
int i;
for (i=2; i<n/2; i++)
{
if (n%i==0)
{
return 0;
}
}
return 1;
}
```

Output

Enter n:7

7 is prime

RESULT

The checking of prime no. was done using User-defined function.

(=) printing prime no. from min to max

```
#include <stdio.h>
int isprime (int, int);
int main ()
{
int min, max, int i;
printf (" Enter min and max:");
scanf ("%d %d", &min, &max);
if (isprime (i)==1)
{
printf ("%d\t", i);
}
}
return 0;
}
int isprime (int n)
{
int i;
for (i=2; i<n/2; i++)
{
if (n%i==0)
{
```

```
return 0;
}
}
return 1;
}
```

Output

Enter min and max 5 20

5 7 11 13 17 19

RESULT

The prime no. from min to max were printed successfully.

PROGRAM #2

Problem analysis

To develop a c-program to find factorial of given number using user defined function with argument & return type.

(Using RECURSIVE function)

ALGORITHM

1. Start
2. Define function prototype
3) Begin main ()
4. Declare if
5. Enter value of n
6. call f: factorial ()
7. print f
8. End main ()
9. Define function factorial (int n)
10. Check if (n==1) then return 1
11. else return (n * factorial (n-1))
12. End function
13. Stop

* C-program

```
# include <stdio.h>
int factorial (int n);
int main ()
{
int n,f;
```

```
plintf ("Enter value of n");
scanf ("%d", &n);
f=factorial (n);
printf ("The factorial of %d is %d", n,f);
return 0;
}
int factorial (int n)
{
if (n==1)
{
return 1;
}
else
{
return (n*factorial (n-1));
}
}
```

Output

Enter value of n:5

The factorial of 5 is 120

RESUIT

The factorial using Recursive function was created successfully.

PROGRAM #5

Problem analysis

ALGORITHM

1. Start
2. Declare prototype
3. Begin main()
4. Declare variables
5. Enter the n and numbers
6. Initialize i=0
7. check if i<n and Read array.
8. i=i+1
9. call avg()
10. print average
12. End main()
13. Define function avg()

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14. Define variables and initialize i=0
15. check if i<n find sum
16. i=i+1
17. avg=sum/n
18. return avg
19. Stop

C-program

```
#include <stdio.h>
float average (int n, int arr[]);
int main()
{
int n, i;
int array [100];
float avg;
printf("Enter the value of n:");
scanf("%d", &n);
printf("Enter %d numbers:\n", n);
for (i=0; i<n; i++)
{
sum ("%d", &arr[i]);
}
avg = average (n, avg);
printf("The average of given numbers is %0.3f", avg);
return 0;
}
float average (int n, int arr[]);
{
int i;
float sum=0, avg;
for (i=0; i<n; i++)
{
sum = sum+arr[i];
}
avg = sum/n;
return avg;
}
```

RESULT

The average of no of array were calculated successfully.

CONCLUSION

From this lab session we learned to implement user define function to perform various task successfully.

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